

Cryptography

CSIT — 5th Semester — 2082 — Answer Sheet
Subject Code: CSC327

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Describe the Fiestel Cipher structure. Given the key {28,7F,,15, 16, 28, AE, D2, A6, AB, F7, 97,66,01,02,03,04}, compute the first 4 byte of next key after first iteration, using the following S-Box in AES.0123456789ABCDEF0637C777BF26B6FC53001672BFED7AB76

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Illustrate the man in middle attack in Diffie - Hellman key exchange protocol. Assume the prime number be 19 and 10 as its primitive root. Select 5 as private key and 4 as random integer. Find the cipher text of M = 2 using Elgamal crypto system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. List the properties of hash function. Describe the algorithm for SHA -1 .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Describe any three types of malicious logic.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Given the key "HELLOWORLD", encrypt the plaintext "TURINGTEST" using Play fair cipher.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain any two modes of block cipher encryption.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. State Fermat's theorem with example. What is the implication of discrete logarithm?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Describe the working mechanism of digital signature algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. List and explain the types of firewall.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Decrypt the ciphertext "HI" using Hill Cipher where the key is 5341.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is digital certificate? Discuss the certificate life cycle.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define authentication system. Discuss about challenge response system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.